Assignment 6

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Title
Development of an E-commerce Platform for Selling Used Items using Node.js, Express.js, and MongoDB

Objectives
The main objectives of this assignment are:

1. To understand the concept of full-stack web application development.
2. To learn backend development using Node.js and Express.js.
3. To integrate MongoDB as a NoSQL database for storing application data.
4. To design an online marketplace for buying and selling used items.
5. To implement a basic user authentication system.

Problem Statement
The goal of this project is to design and develop an E-commerce web application for used item sales using Node.js and Express framework.
The application should allow users to:

- Register and log in securely
- Post their used items for sale
- Browse available products
- View seller details and contact them

All user data, product details, and interaction information should be stored in a MongoDB database to ensure proper data management and retrieval.

Outcomes
After completing this assignment, the following outcomes are achieved:

1. Ability to develop backend systems using Node.js.
2. Understanding of routing and middleware using Express.js.
3. Experience in working with MongoDB for data storage.
4. Implementation of a user login and authentication system.
5. Development of a real-world marketplace application.

Hardware and Software Requirements
Hardware Requirements

- Computer or Laptop
- Minimum 4GB RAM

Software Requirements

- Node.js
- MongoDB
- Visual Studio Code
- Web Browser (Chrome / Firefox)

Theory

1. Node.js
Node.js is a runtime environment that enables execution of JavaScript on the server side. It is widely used for building scalable applications due to its asynchronous and non-blocking nature.

2. Express.js
Express.js is a lightweight framework that simplifies server-side development. It helps in managing routes, handling requests, and building APIs efficiently.

3. MongoDB
MongoDB is a NoSQL database that stores data in a flexible document format. It is suitable for applications where data structure may vary, such as e-commerce platforms.

4. Mongoose
Mongoose is an ODM library used with MongoDB. It provides schema-based modeling, validation, and simplifies database operations.

5. E-commerce System for Used Products
This system acts as a marketplace where users can:
- List their used items for sale
- Browse items posted by others
- Explore different categories such as vehicles, electronics, and furniture

6. User Authentication System
The application includes a login system to ensure security:
- Users must register before accessing features
- Login is required to post items
- Each product is linked to its seller

7. Buyer and Seller Interaction
The system allows interaction between buyers and sellers:
- Sellers provide product details and contact information
- Buyers can view listings
- Buyers contact sellers directly to complete the deal

8. Working of the Application
The working process of the system is as follows:
1. The user visits the website.
2. The user registers or logs in.
3. Logged-in users can add products for sale.
4. Product details are stored in the database.
5. Other users browse available listings.
6. A buyer selects a product.
7. The buyer contacts the seller.
8. The transaction is completed outside the system.

Flowchart

Start
↓
User visits website
↓
Register / Login
↓
Add product for sale
↓
Store data in MongoDB
↓
Display products
↓
Buyer selects product
↓
Contact seller
↓
End

Design / Execution Steps

The following steps were followed during development:

1. Installed Node.js and MongoDB.
2. Created a project directory.
3. Initialized the project using npm.
4. Installed required dependencies like Express and Mongoose.
5. Built the backend server using Express.js.
6. Connected the application with MongoDB.
7. Designed schemas for users and products.
8. Implemented user registration and login functionality.
9. Developed frontend using HTML, CSS, and JavaScript.
10. Added functionality for posting and viewing products.
11. Enabled seller contact feature.
12. Integrated frontend and backend using API routes.
13. Tested the application in a browser.

Test Cases

Test Case	Description	Expected Result
Server Start	Run application	Server starts successfully
User Login	Enter valid credentials	Login successful
Add Product	Submit product details	Product stored in database
View Products	Open homepage	Products displayed
Contact Seller	Select product	Seller details visible

Code:

JavaScript:

```
const express = require("express");
const mongoose = require("mongoose");
const session = require("express-session");

const app = express();
const PORT = 3000;

// Middleware
app.use(express.urlencoded({ extended: true }));
app.use(express.json());

app.use(session({
   secret: "used-items-secret",
   resave: false,
   saveUninitialized: true
}));

// DB Connection
mongoose.connect("mongodb://127.0.0.1:27017/usedItemsDB")
.then(() => console.log("MongoDB Connected"))
.catch(err => console.log(err));

app.listen(PORT, () => {
   console.log(`Server running on port ${PORT}`);
});
```

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Used Treasure Hub</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <!-- Navbar -->
  <header class="navbar">
    <h1 class="logo">UsedTreasure</h1>
    <input type="text" placeholder="Search items..." class="search-bar">
    <button class="btn">Login</button>
  </header>

  <!-- Hero Section -->
  <section class="hero">
    <h2>Buy & Sell Used Items Near You</h2>
    <p>Find great deals on cars, electronics, furniture and more.</p>
    <button class="primary-btn">Browse Listings</button>
  </section>

  <!-- Product Section -->
  <section class="products">

    <div class="card">
      <img src="https://via.placeholder.com/250" alt="product">
      <h3>iPhone 14 Pro</h3>
      <p class="price">₹72,000</p>
      <p class="location">Pune, Maharashtra</p>
      <button class="contact-btn">Contact Seller</button>
    </div>

    <div class="card">
      <img src="https://via.placeholder.com/250" alt="product">
      <h3>Royal Enfield Classic 350</h3>
      <p class="price">₹1,45,000</p>
      <p class="location">Mumbai, Maharashtra</p>
      <button class="contact-btn">Contact Seller</button>
    </div>
```

```
 <div class="card">
 <img src="https://via.placeholder.com/250" alt="product">
 <h3>Dell XPS Laptop</h3>
 <p class="price">₹68,000</p>
 <p class="location">Bangalore, Karnataka</p>
 <button class="contact-btn">Contact Seller</button>
 </div>

 </section>

</body>
</html>
```

CSS

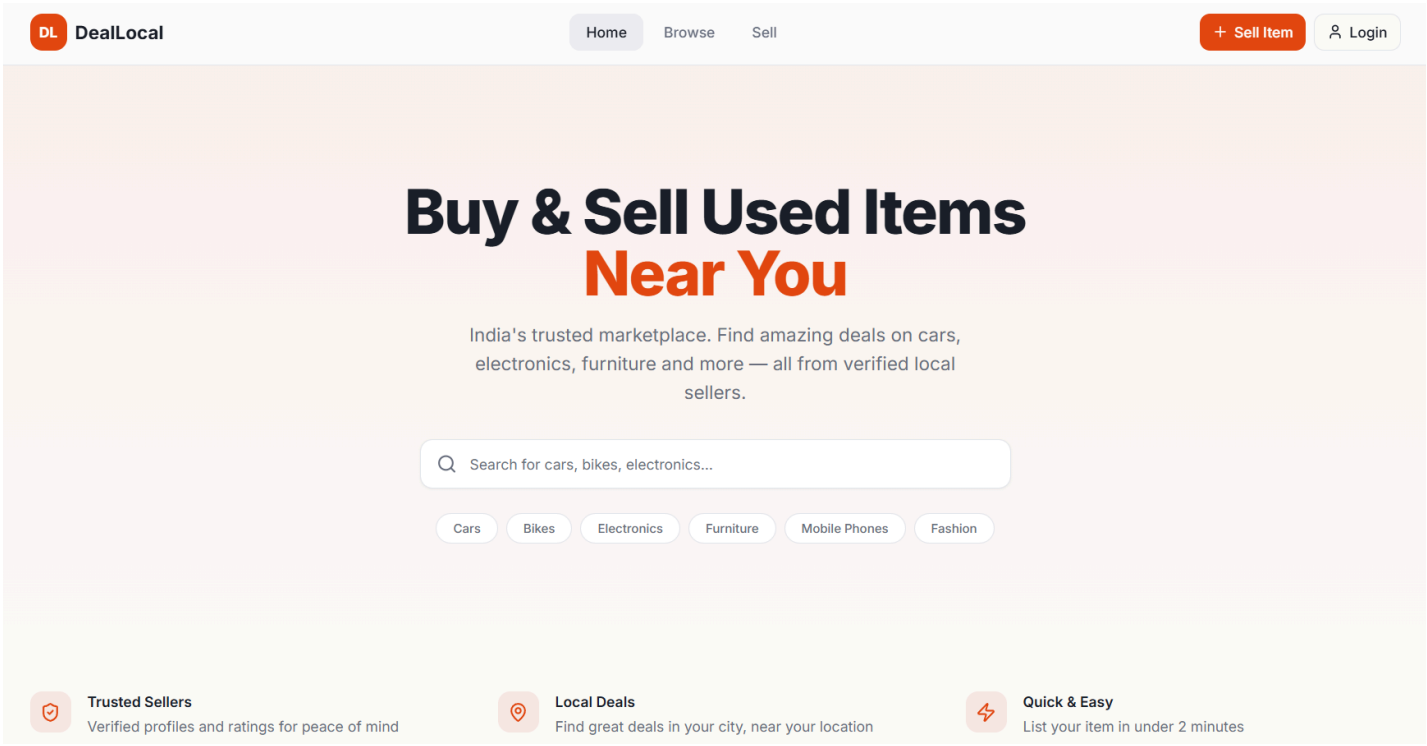
```
body {
    font-family: 'Poppins', sans-serif;
    margin: 0;
    background-color: #f5f5f5;
}

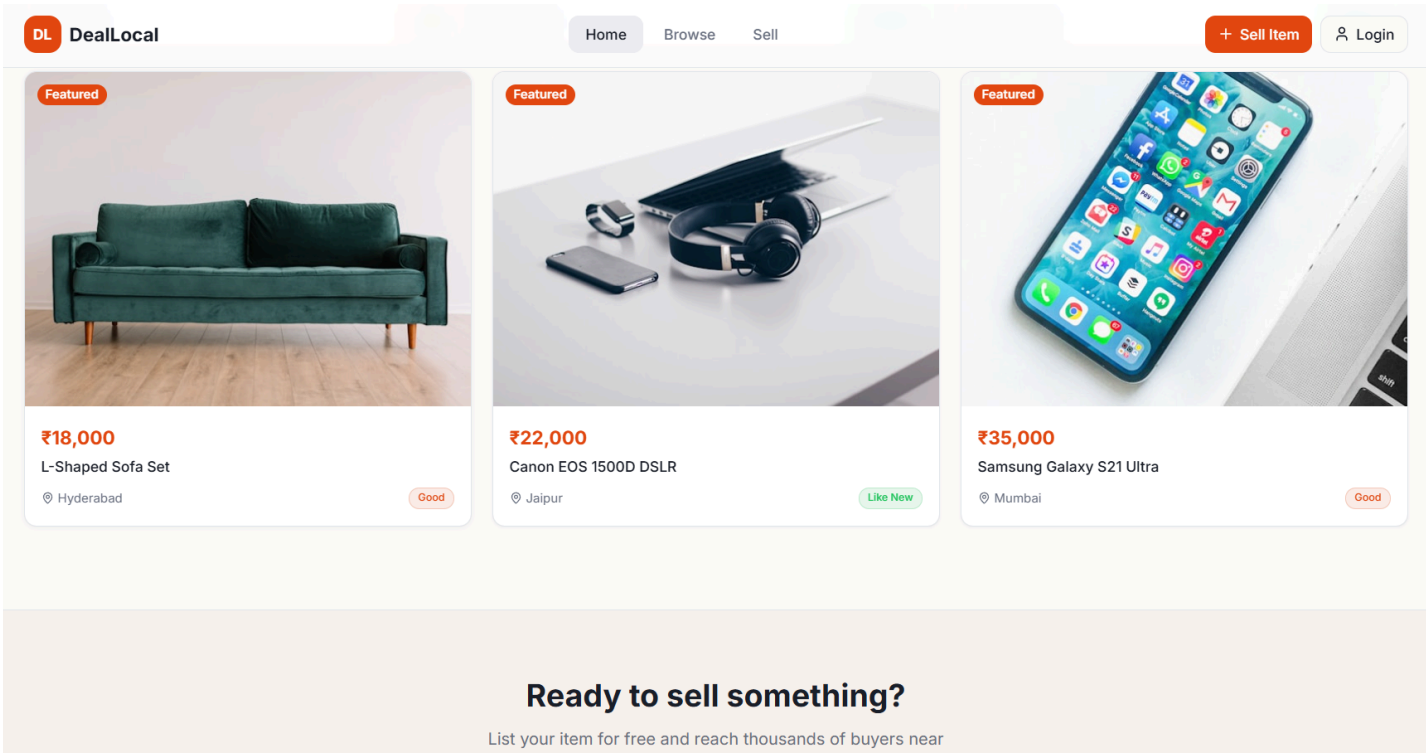
/* Navbar */
.navbar {
    display: flex;
    justify-content: space-between;
    align-items: center;
    padding: 15px 30px;
    background-color: #1f1f1f;
    color: white;
}

.logo {
    font-size: 22px;
    font-weight: bold;
}

.search-bar {
    width: 40%;
    padding: 8px;
    border-radius: 5px;
    border: none;
}
```

Output:


DL
DealLocal
Home
Browse
Sell
+ Sell Item
Login
Buy & Sell Used Items
Near You
India's trusted marketplace. Find amazing deals on cars, electronics, furniture and more — all from verified local sellers.
Search for cars, bikes, electronics...
Cars
Bikes
Electronics
Furniture
Mobile Phones
Fashion
Trusted Sellers
Verified profiles and ratings for peace of mind
Local Deals
Find great deals in your city, near your location
Quick & Easy
List your item in under 2 minutes


DL
DealLocal
Home
Browse
Sell
+ Sell Item
Login
Featured
₹18,000
L-Shaped Sofa Set
Hyderabad
Good
Featured
₹22,000
Canon EOS 1500D DSLR
Jaipur
Like New
Featured
₹35,000
Samsung Galaxy S21 Ultra
Mumbai
Good
Ready to sell something?
List your item for free and reach thousands of buyers near

DL DealLocal

Home Browse Sell

+ Sell Item Login

Sell an Item

Fill in the details below to list your item.

Product Title

e.g. iPhone 13 Pro 128GB

Category

Select category

Price (₹)

25000

Condition

Select condition

Location

e.g. Mumbai

Description

Describe your item...

Upload Image

Click to upload or drag & drop

Post Your Ad

DL DealLocal

Home Browse Sell

+ Sell Item Login

Mobile Phones

2 listings found

Search for cars, bikes, electronics...

Filters 1 | Cars | Bikes | Electronics | Furniture | Mobile Phones | Fashion | Books | Sports | Home Appliances | Others



₹42,000

iPhone 13 Pro 128GB

Chennai Like New



Featured

₹35,000

Samsung Galaxy S21 Ultra

Mumbai Good

DL DealLocal

India's trusted marketplace for buying and

Quick Links

Browse Products

Categories

Cars

Support

Help Center

Conclusion

In this assignment, an E-commerce platform for selling used items was successfully developed using Node.js, Express.js, and MongoDB. The application provides features such as user authentication, product listing, and direct interaction between buyers and sellers.

This project demonstrates how full-stack technologies can be combined to build a functional marketplace system, giving practical exposure to real-world application development.